Structural Sparsity for Quantum Simulation: A Synthesis of the GeoVac Quantum-Computing Arc*

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The Geometric Vacuum (GeoVac) framework discretizes electronic structure on natural geometries, producing sparse graph Hamiltonians whose angular content is fixed by Gaunt selection rules. This synthesis assembles the framework's quantum-computing arc—the four Group 4 papers—into a single resource-oriented narrative. The spine is: the qubit encoding and its Pauli/1-norm scaling [2]; the benchmarked resource library across three periodic-table rows [4]; the S_N representation-theoretic foundation that classifies the atomic building blocks [3]; and the nuclear shell-model extension that delineates exactly which part of the framework transfers [6]. The unifying value proposition is *structural*, not accuracy-based: after the Jordan–Wigner transformation the GeoVac lattice Hamiltonian yields $\sim Q^{3.8}$ Pauli terms for atoms (corrected 2026-08-29; the retired pair-diagonal encoding gave $O(Q^{3.15})$) and a Pauli count exactly linear in qubit count for composed multi-center molecules at fixed basis (within-molecule basis exponent 3.17; corrected 2026-08-29 from the retired $O(Q^{2.5})$) (versus $O(Q^{3.9–4.3})$ for Gaussian baselines), with a sub-quadratic Pauli 1-norm $\lambda = O(Q^{1.77})$ ($R^2 = 0.9997$; re-measured 2026-08-30, retired: $O(Q^{1.69})$) that controls Trotter depth and qubitization query cost. Two disciplines run through the arc. *First*, we separate what survived the 2026-08-29 selection-rule correction from what did not. *Surviving:* the exact linearity $N_{\text{Pauli}} = c\,Q$ across molecules at fixed basis, the isostructural invariance, the Z-independence, and the equal-qubit 1-norm advantage (3.8× at $Q = 10$, 7.1× at $Q = 28$) — each re-measured under the exact rule and re-confirmed, not inherited. *Not surviving:* the within-molecule and atomic exponents (which moved 2.5 → 3.17 and 3.15 → 3.8), the QWC exponent (which moved 3.36 → 4.01, above the term exponent, withdrawing the measurement-group advantage entirely), and every absolute multiplier. The earlier framing of this discipline — "scaling exponents unchanged, only multipliers re-price" — was itself retracted with the correction: the re-pricing is not a constant factor, it grows along the basis axis. *Second*, every comparison is made at *matched qubit count*, not matched accuracy: the composed basis carries classical equilibrium-geometry errors of 5–26%, so the framework is positioned for fixed-geometry NISQ/VQE simulation (Pauli count, QWC measurement groups), not for fault-tolerant geometry optimisation. The nuclear extension is honest in the same way: its binding energies are encoding-validation benchmarks far from experiment, and the Fock projection rigidity theorem states precisely that the conformal S^3 structure does *not* transfer to the harmonic-oscillator basis—only the angular machinery and the qubit encoding do.

I. INTRODUCTION: A STRUCTURAL VALUE PROPOSITION

The classical-solver investigation that occupies the GeoVac quantum-chemistry arc (Group 2) reached a clear verdict: the natural-geometry hierarchy characterises what discrete graph Hamiltonians can and cannot achieve for ground-state potential-energy surfaces, and on the accuracy axis it is not competitive with production quantum chemistry. The framework's computational value lies elsewhere. The same construction that makes the classical accuracy bounded—a single-center angular basis with Gaunt-enforced selection rules and a pseudopotential-partitioned composed geometry—makes the resulting *qubit* Hamiltonians structurally sparse. This synthesis collects the four papers that develop and benchmark that observation.

The reader should hold two framings in mind throughout. First, the advantage is in *resource scaling and term counts*, measured at *matched qubit count*, not in energy accuracy: at $n_{\max} = 2$ the composed basis carries 5–26% classical equilibrium-geometry errors ($LiH/BeH_2/H_2O$), so the natural use is single-point simulation at a fixed nuclear geometry, where the sparsity is free and the geometry error does not enter. Second, the framework targets the NISQ/VQE regime—where Pauli term count and the number of qubitwise-commuting (QWC) measurement groups dominate cost—rather than the fault-tolerant QPE regime, although the sub-quadratic Pauli 1-norm is also the relevant quantity for qubitization.

The four papers form a dependency spine. Paper 14 [2] establishes the encoding and its scaling laws. Paper 20 [4] benchmarks a 37-system library (35 composed molecules + He + H_2) across three periodic-table rows and exports them to the standard quantum-software ecosystems. Paper 16 [3] supplies the S_N representation-theoretic classification of the atomic building blocks that the composed builder assembles. Paper 23 [6] extends the angular machinery to the nuclear shell model and proves, via the Fock rigidity theorem, the boundary of what transfers.

* Group 4 synthesis in the Geometric Vacuum series

II. THE ANGULAR-SPARSITY MECHANISM

The structural origin of every resource result in this arc is a single fact about the two-electron integral (ERI) tensor. In a single-center basis indexed by (n, ℓ, m), the angular part of each ERI is a product of Gaunt coefficients, and a Gaunt coefficient vanishes unless its three angular momenta satisfy the $3j$ triangle inequality and the magnetic selection rule. As the number of spatial orbitals M grows, the fraction of nonzero ERIs therefore *decays*—empirically as $\approx M^{-0.49}$ (corrected 2026-08-29; the retired rule's $M^{-0.85}$ overstated the decay)—whereas a Gaussian ERI tensor remains near-dense at $O(1)$ occupancy. This is the potential-independent angular-sparsity theorem of Paper 22 (Group 3): the ERI density depends only on $\ell_{\max}$, not on the radial potential $V(r)$. The sparsity survives the Jordan–Wigner transformation because it is carried by the angular labels, which JW does not mix.

a. The selection rule, and a correction to it. The exact Coulomb angular selection rule is the global-M_L rule, $m_a + m_b = m_c + m_d$, and every number in this synthesis is now computed under it.

Until 2026-08-29 the production composed and atomic builders realised instead a *pair-diagonal* restriction $m_a = m_c \wedge m_b = m_d$, described in earlier versions of this document as a deliberate sparsifying approximation. An independent re-derivation traced it to a *sign error*: the Gaunt coefficient set the Wigner-3j magnetic argument to $q = m_c - m_a$ where the bottom row requires $q = m_a - m_c$, so the coefficient vanished identically unless $m_a = m_c$, silently deleting every m-changing multipole (88 of the 126 nonzero c^k at $l \leq 2$) — among them the genuinely nonzero "m-swap" integrals (e.g. $\langle p_{+1}p_{-1}|p_{-1}p_{+1}\rangle$, angular factor $\sqrt{6}/5 \approx 0.49$). The error was arbitrated against direct quadrature of the angular factor from its definition.

Two consequences, both of which retract earlier statements here:

- The re-pricing is *not* a constant factor. It is $2.51\times$ uniformly across main-group molecules at $n_{\max} = 2$, but grows along the basis axis ($1.00\times/2.51\times/5.40\times$ at $n_{\max} = 1/2/3$), so the within-molecule exponent moves $2.54 \to 3.17$. The earlier "constant factor, scaling unchanged" claim was a single-point artifact.

- Two defects were corrected together and they behave oppositely, which is worth separating because the distinction was previously collapsed. The wrong-sign Gaunt argument ($q = m_c - m_a$) deleted every m-changing multipole and so changed the ERI *support*: $65 \to 107$ nonzero entries per block, which is why every count, density and exponent moved. The Condon–Shortley factor-order error, by contrast, leaves the ERI support exactly invariant and flips signs only — measured on the relativistic block as identical support (2,676 entries) and identical $\sum|\mathrm{ERI}|$ with 40.4% of signs reversed. It is not resource-neutral even so: the sign flips change which Jordan–Wigner terms cancel, moving $\mathrm{LiH_{rel}}$ from 1,413 to 1,501 Pauli terms. The distinction is between the ERI level, where only the sign error moved the support, and the resource level, where both defects moved counts. "Values, not entries" is true of the second and false of the first; an earlier version of this synthesis generalised it to the whole correction and thereby claimed the counts were invariant when they were not. What survives is the *form* $N_{\mathrm{Pauli}} = c\,Q$, with its coefficient re-measured ($11.10 \to 27.90$).

Every resource number reported here is the exact global-M_L value; retired pair-diagonal figures appear only where explicitly marked as retired history. There is no dual-rule convention to adopt: the pair-diagonal rule was a wrong-sign Gaunt argument, not an alternative to the exact one.

III. THE ENCODING AND ITS SCALING LAWS (PAPER 14)

Paper 14 transforms the GeoVac lattice Hamiltonian to qubits via Jordan–Wigner and characterises the resulting Pauli decomposition by several operationally distinct metrics.

a. Term count. For single-center atoms the Pauli term count scales as $\sim Q^{3.8}$ in the qubit number Q (fit 3.78; corrected 2026-08-29 from the retired $O(Q^{3.15})$), which sits at the lower edge of — not below — the Gaussian molecular Pauli-scaling exponents in the $Q^{3.9}$–$Q^{4.3}$ range; the specific electronic-structure values $Q^{4.25}$ (LiH) and $Q^{3.92}$ (H_2O), and the range they bracket, are GeoVac's own log-log fit of the published Gaussian Pauli counts of Trenev et al. [7] (Table 5, Appendix B; Jordan–Wigner with 2-qubit reduction). For composed multi-center molecules the pseudopotential-partitioned architecture produces block-diagonal ERIs and a *universal* within-molecule exponent of 3.17 (identical to all reported digits — corrected 2026-08-29 from the retired $O(Q^{2.5})$, whose apparent 0.02 spread was an artifact of the sign error), across the benchmarked molecules—independent of molecular topology. Across the library the composed count is, remarkably, *exactly linear* in the qubit count,

$$N_{\mathrm{Pauli}} = c\,Q, \qquad c = 27.90 \text{ (main group)}, \quad 30.03 \text{ (d-block)}, \tag{1}$$

under the exact global-M_L rule at $n_{\max} = 2$. (The retired pair-diagonal rule gave 11.10 and 9.23; note that the ordering *reverses* — the d-block, which appeared *sparser* under the retired rule, is in fact *denser*, because higher angular momentum carries more m-swap channels for the sign error to drop. The pair-diagonal d-block sparsity was an artifact, not a selection-rule property.)

b. Measurement and Trotter cost. Two further metrics govern practical cost, and the 2026-08-30 remeasurement separates them sharply. The number of QWC measurement groups—the dominant factor in the measurement overhead of variational algorithms—scales as $O(Q^{4.01})$, which is *steeper* than the Pauli term exponent (3.77) and lies inside the Gaussian $O(Q^{4-5})$ band. No measurement-group scaling advantage is therefore claimed; under the retired pair-diagonal rule this exponent read $O(Q^{3.36})$, below the term count, and that reading is withdrawn. The Pauli 1-norm $\lambda = \sum_i |c_i|$, which bounds both Trotter step counts and qubitization query complexity, scales as $O(Q^{1.77})$ with $R^2 = 0.9997$ (retired: $O(Q^{1.69})$). This sub-quadratic 1-norm scaling survives the correction and remains the key fault-tolerant-relevant result: Trotter circuit depth grows far more slowly with system size than the raw term count alone would suggest. The *molecular* Gaussian baselines (LiH, H_2O) are the published Pauli counts of Trenev et al. [7] (Table 5, Appendix B), cross-checked against analytic references (Szabo and Ostlund for H_2; analytical Slater-type orbitals for He). The 6-31G and cc-pVDZ entries of Paper 14's Table I use synthetically generated integrals and are retained for scaling-fit purposes only.

c. Tier discipline. The scaling exponents are *measured* quantities (log-log fits with reported R^2), not derived; the angular selection rules underlying the ERI-density decay are *exact*; and the composed sparsity advantage *magnitude* is a measured quantity under a named ERI rule. The construction itself is zero-parameter: the Hamiltonian follows from nuclear charges and geometry alone.

IV. THE BENCHMARKED RESOURCE LIBRARY (PAPER 20)

Paper 20 turns the encoding into a usable artifact: the `hamiltonian()` registry of 37 systems (35 composed molecular qubit Hamiltonians spanning $Z = 1$–56 (H through Ba), plus the He and H_2 reference atoms), including multi-center systems (CO, N_2, F_2), mixed frozen-core molecules (NaCl), and the first transition series as hydrides (ScH through ZnH), exported to Qiskit, OpenFermion, and PennyLane.

a. Two architectures. The library exposes two composed-molecular architectures with complementary strengths. The *composed* (pseudopotential, PK) architecture is the sparser of the two: LiH requires 838 Pauli terms at 30 qubits, with 4.3× fewer QWC groups (64 vs. 273) than the Gaussian STO-3G reference at 12 qubits (raw baseline). The small-system market test is near parity on the raw-vs-raw comparison — 838 versus STO-3G's 907 (1.08× fewer) — and *unfavourable* against the qubit-reduced count (276, i.e. GeoVac is 3.0× larger). (Corrected 2026-08-29; the retired pair-diagonal figures were 334 Pauli and 13× QWC.) The sparsity advantage spans 0.28× (H_2O against minimal-basis STO-3G, where GeoVac is *denser*) through 55–76× against cc-pVDZ to 317× at the most favourable equal-qubit point; the retired pair-diagonal rule supported a blanket "one to three orders of magnitude" that survives at neither end. What advantage there is rests on the scaling gap and the multi-center block structure, *not* on the small-molecule minimal-basis count—which is near parity. The *balanced coupled* (PK-free) architecture trades sparsity for chemistry: it binds LiH with an FCI minimum at a computed $R_{\mathrm{eq}} = 3.227$ bohr (7.0% above the experimental 3.015 bohr) at $n_{\max} = 2$ (2,726 Pauli terms, 30 qubits; corrected 2026-08-29 from the retired 878); a single-point evaluation at the experimental geometry reaches a 0.20% energy error at $n_{\max} = 3$ when solved exactly (number-projected FCI).

b. Row-conditional chemistry scope. The chemistry-accuracy scope at $n_{\max} = 2$ is honestly *row-conditional*: first-row hydrides (LiH, BeH_2, H_2O, ...) bind with documented PES topology, while second-row hydrides (NaH and below) exhibit monotone overattraction under every tested configuration—frozen-core balanced, multi-zeta valence, screened cross-center, explicit-core Hartree–Fock—with no interior PES minimum at the qubit-Hamiltonian level. This scope boundary is documented empirically, not papered over: the framework's value proposition is structural sparsity with a proven state-space Gromov–Hausdorff truncation rate (Paper 38) — a *metric-space* rate, explicitly *not* a direct energy-error bound (the operator-to-energy lift overshoots by 3–4 orders of magnitude) — not energy-accuracy parity with CCSD(T) on second-row systems.

c. Frozen cores. Second- and third-row molecules use analytical frozen cores ([Ne], [Ar], [Ar]$3d^{10}$); the frozen-core energy enters only through the identity operator and does not affect the quantum measurement cost. Isostructural molecules with frozen cores produce identical Pauli counts across rows—the periodic-table structure is visible directly in the resource counts.

V. THE REPRESENTATION-THEORETIC FOUNDATION (PAPER 16)

The composed builder assembles molecules from atomic building blocks, and Paper 16 supplies the group-theoretic classification of those blocks. After separation of the hyperradius, the N-electron angular dynamics live on S^{3N-1} and the angular kinetic energy is the SO($3N$) Casimir operator with eigenvalue $\mu_{\mathrm{free}} = \nu(\nu+3N-2)/2$. Fermionic antisymmetry restricts the spatial wavefunction to the S_N irreducible representation conjugate to the spin Young diagram, whose first-column length λ_1 fixes the minimum angular quantum number $\nu = N - \lambda_1$. For every atom with ground-state spin $S < N/2$ the spatial irrep has $\lambda_1 = 2$, giving the universal result

$$\nu = N - 2, \qquad \mu_{\mathrm{free}} = 2(N-2)(N-1). \qquad (2)$$

This produces a five-type classification of atoms—A (trivial), B (noble gas), C (alkali), D (alkaline earth), E (open shell)—and recasts the periodic law as the repetition of the S_N irrep sequence C → D → E → B within each period. The classification is implemented in `atomic_classifier.py`, which refines the open-shell type E into a p-block dispatch (E) and a separate d-block dispatch (F) for the composed builder, and supplies the universal angular quantum number $\nu = N - 2$.

a. Honest framing. The classification *maps onto, rather than predicts*, the known periodic table: its value is a unified group-theoretic language for a pattern usually described empirically, not a derivation of chemistry from first principles. Paper 16 also establishes a structural fact relevant to the relativistic extension: the topology (ν, μ_{free}) is smooth through $Z = 1/\alpha \approx 137$, so the Dirac instability is a *metric* singularity on S^{3N-1}, not a topological one.

VI. THE NUCLEAR EXTENSION AND ITS BOUNDARY (PAPER 23)

Paper 23 applies the angular-sparsity machinery to a system the conformal S^3 projection cannot reach—the nuclear shell model—and in doing so delineates exactly which part of the framework is universal.

a. What the angular machinery delivers. Built on the harmonic-oscillator single-particle basis with a Mayer–Jensen spin–orbit term, the construction reproduces the HO shell closures $2, 8, 20, 40, 70, 112$ directly from graph state counting, and recovers the physical magic numbers $2, 8, 20, 28, 50, 82, 126$ with a spin–orbit strength $v_{ls}/\hbar\omega \approx 0.17$ and a Nilsson $\ell(\ell+1)$ correction $d_{\ell\ell}/\hbar\omega \approx 0.02$. The qubit encodings inherit the angular sparsity: the deuteron $(1p + 1n)$ requires 16 qubits and 688 non-identity Pauli terms using the Minnesota nucleon–nucleon potential and Moshinsky–Talmi brackets; helium-4 $(2p + 2n)$ requires 16 qubits and 828 Pauli terms despite a Hilbert space 12.25× larger—only 1.20× more terms, a direct consequence of the angular selection rules carrying through to multi-particle systems. A composed nuclear–electronic deuterium encoding (26 qubits) validates the multi-block architecture and quantifies the $\sim 10^{13}$ coefficient ratio between nuclear, electronic, and hyperfine scales.

b. The rigidity boundary.

Theorem 1 (Fock projection rigidity)**.** *The S^3 conformal projection that makes electronic Hamiltonians equivalent to a graph Laplacian is unique to the Coulomb potential and does not extend to the harmonic-oscillator or Woods–Saxon systems.*

For nuclear Hamiltonians the GeoVac contribution is therefore the angular machinery and qubit encoding only, *not* the conformal projection. This is the structural dual of the foundations-side HO rigidity theorem (Paper 24).

c. Honest framing. All nuclear physics inputs (Minnesota parameters, spin–orbit strength, $\hbar\omega$) follow the standard shell-model literature—they are not GeoVac-derived—and the quantitative binding energies at $N_{\text{shells}} = 2$ are far from experiment. They are to be read as *encoding-validation benchmarks*, demonstrating that the two-species sparse qubit architecture works, not as a nuclear-structure calculation.

VII. HONEST SCOPE AND POSITIONING

a. What is robust. What survives the correction is *structure*, not numbers. The exact linearity $N_{\text{Pauli}} = c\,Q$ across molecules at fixed basis (1), the isostructural invariance, the Z-independence of the ERI count, and the block-diagonal selection-rule sparsity all hold under the exact rule, and rest on the potential-independent angular-sparsity theorem (Paper 22). Indeed the linearity is *cleaner* after the correction: the two-point within-molecule exponent is now identical across all six molecules and forced rather than fitted.

Every *exponent*, by contrast, moved: atomic 3.15 → 3.8, within-molecule 2.5 → 3.17, 1-norm 1.69 → 1.774, QWC 3.36 → 4.013 — and the QWC exponent moved far enough to overturn its own conclusion, since 4.013 exceeds the Pauli exponent and lies inside the Gaussian $O(Q^{4\text{–}5})$ band. An earlier version of this paragraph listed those exponents under "robust" while stating two sentences later that they had moved. The qubit library is a concrete, exported, reproducible artifact.

b. What is conditional. The *absolute* sparsity multipliers and small-system market-test lines were re-priced on 2026-08-29 when the pair-diagonal ERI restriction was found to be a sign error; the re-pricing is basis-dependent, not a constant factor, and it leaves composed LiH at near parity with raw STO-3G (838 vs. 907) and 3.0× larger than the qubit-reduced count. All Pauli comparisons are at matched qubit count, not matched accuracy: the composed basis carries 5–26% classical R_{eq} errors, the chemistry-accuracy scope is row-conditional (first-row binds; second-row overattracts), and the nuclear binding energies are encoding-validation benchmarks far from experiment.

c. What was tried and did not work. The arc includes several honest negatives that bound the design space. Discrete $\mathbb{Z}_2$ tapering beyond the Hopf $m_\ell \to -m_\ell$ symmetry does not transfer to relativistic chemistry: neither κ-parity, direct-basis m_j-parity, nor rotated-basis m_j-parity commutes with the Dirac–Coulomb Hamiltonian, because the jj-coupled Coulomb operator couples sign-counted sectors under sum-conservation (not parity) rules. Adding off-diagonal cross-block h_1 coupling to gain binding produces 16× over-binding on LiH; the framework's binding wall — the structural obstruction to chemistry-accuracy binding documented in the Group 2 record — is structurally inseparable from its encoding advantages. A non-abelian gauge upgrade of the composed Dirac does not reduce the Pauli count below the $\mathbb{Z}_2$-tapered identity, because tapering requires a Hamiltonian *symmetry* and the gauge freedom is a basis change, not a symmetry. The spectral action is the wrong functional for chemistry observables at finite cutoff. These negatives are recorded so they are not re-derived.

d. Positioning. The natural target is variational, NISQ-era simulation—where the Pauli term count and the number of QWC measurement groups dominate cost—at fixed molecular geometry. The sub-quadratic Pauli 1-norm makes the encodings relevant to qubitization as well, but the framework is not positioned as a fault-tolerant-QPE or geometry-optimisation tool, and it is not a replacement for production quantum chemistry. It is a structurally sparse, zero-parameter, ecosystem-exported library of qubit Hamiltonians with a proven state-space Gromov–Hausdorff truncation rate (a metric-space rate, not an energy-error bound)—a research instrument for the resource-estimation and algorithm-development side of quantum simulation.

[1] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation from Scale-Invariant Graph Topology," GeoVac Paper 7 (2026).

[2] J. Loutey, "Structurally Sparse Qubit Hamiltonians from Spectral Graph Theory," GeoVac Paper 14 (2026).

[3] J. Loutey, "The Periodic Table Recast as S_N Representation Theory on S^{3N-1}," GeoVac Paper 16 (2026).

[4] J. Loutey, "Structurally Sparse Molecular Hamiltonians for Near-Term Quantum Simulation," GeoVac Paper 20 (2026).

[5] J. Loutey, "Potential-Independent Angular Sparsity for Qubit Hamiltonians in Spherical Fermion Systems," GeoVac Paper 22 (2026).

[6] J. Loutey, "Nuclear Shell Model Hamiltonians on the Hyperspherical Lattice," GeoVac Paper 23 (2026).

[7] D. Trenev, P. J. Ollitrault, S. M. Harwood, T. P. Gujarati, S. Raman, A. Mezzacapo, and S. Mostame, "Refining resource estimation for the quantum computation of vibrational molecular spectra through Trotter error analysis," Quantum **9**, 1630 (2025); arXiv:2311.03719. [Cited for the electronic-structure Gaussian-basis Pauli counts in Table 5 (Appendix B; Jordan–Wigner with 2-qubit reduction), the Gaussian baseline; the $Q^{3.9}$–$Q^{4.3}$ scaling exponents (and the specific LiH/H_2O exponents) are GeoVac's own log-log fit of those published counts, per Paper 14.]